Experiment-15

Write A Python Program To Implement Decision Tree

Aim

To write a Python program to implement a Decision Tree for predicting whether a student will pass or fail based on study hours and attendance.

Algorithm

1. Start the program.
2. Initialize the training data with study hours and attendance.
3. Assign the output labels as Pass or Fail.
4. Define the decision tree rules.
5. Get the student's study hours and attendance.
6. Check whether the study hours are less than or equal to 3.
7. If true, predict Fail.
8. Otherwise, check whether attendance is less than or equal to 70.
9. If true, predict Fail; otherwise, predict Pass.
10. Display the predicted result.
11. Stop the program.

Code

```
# Decision Tree implementation

# Training data: [Study Hours, Attendance]
X = [
    [2, 60],
    [3, 65],
    [5, 80],
    [6, 90],
    [1, 40],
    [7, 95]
]

# 0 = Fail, 1 = Pass
y = [0, 0, 1, 1, 0, 1]

# Decision tree rules
def predict(study_hours, attendance):
    if study_hours <= 3:
        return "Fail"
    elif attendance <= 70:
        return "Fail"
    else:
        return "Pass"

# Get student details
hours = 4
attendance = 75

# Predict result
result = predict(hours, attendance)

print("Study Hours:", hours)
print("Attendance:", attendance)
print("Predicted Result:", result)
```

Output

```
>>>
============================== RESTART: D:/ex-15.py ==============================
Study Hours: 4
Attendance: 75
Predicted Result: Pass
>>>
```

Result

The Decision Tree program was implemented successfully and predicted the student's result based on study hours and attendance.